MACHINE LEARNING-BASED ICU MORTALITY RISK PREDICTION USING CLINICAL DATA

A Recall-Focused and Robust Approach for Early Identification of High-Risk Patients

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Introduction

This study focuses on predicting mortality risk in Intensive Care Unit (ICU) patients using machine learning techniques. Early identification of high-risk patients is critical for timely clinical intervention and improved patient outcomes. Traditional scoring systems such as APACHE and SOFA provide useful insights but may not capture complex relationships within clinical data. This project addresses these limitations by applying data-driven modeling while considering real-world challenges such as class imbalance and data noise.

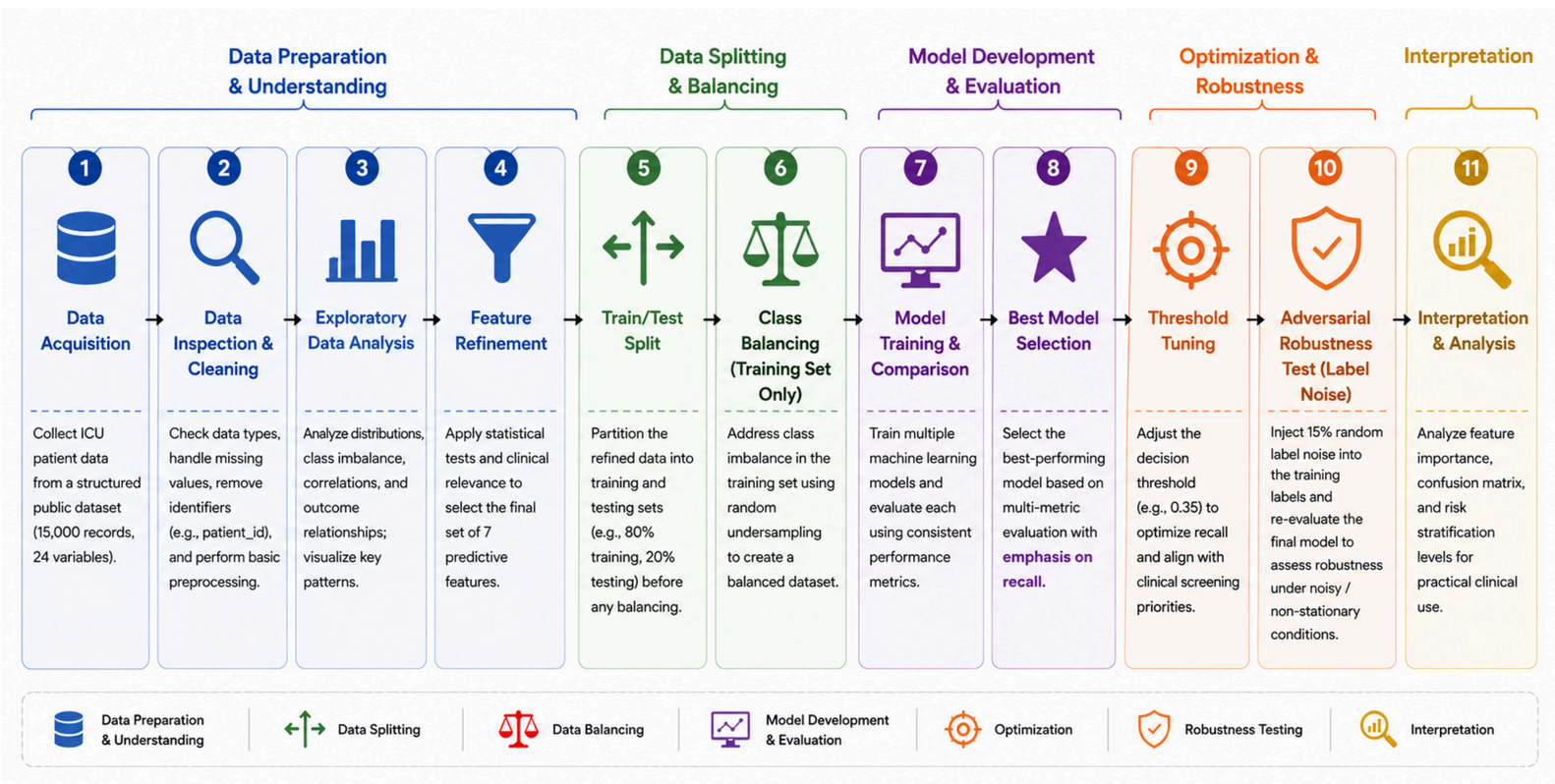
Objective

The objective of this project is to develop a machine learning model that accurately identifies high-risk ICU patients. The system prioritizes recall to minimize missed critical cases while maintaining balanced overall performance. Additionally, the project aims to evaluate model robustness under imperfect data conditions and provide a practical decision-support tool for healthcare environments.

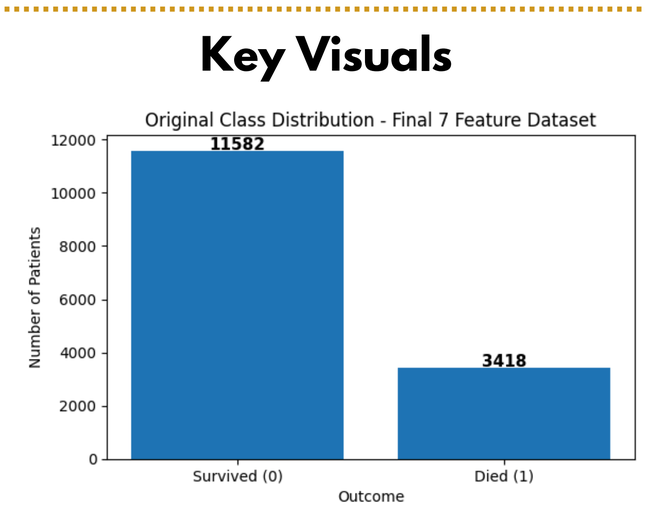
Dataset

- 15,000 ICU patient records
- 24 features → reduced to 7
- Target: mortality (0 = survived, 1 = died)
- Imbalanced dataset:
  - Survived: 11,582
  - Died: 3,418
- It includes clinical and physiological features

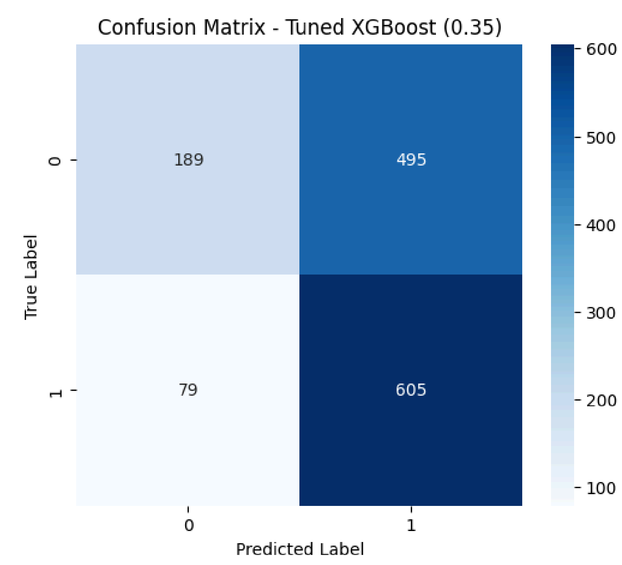
Methodology



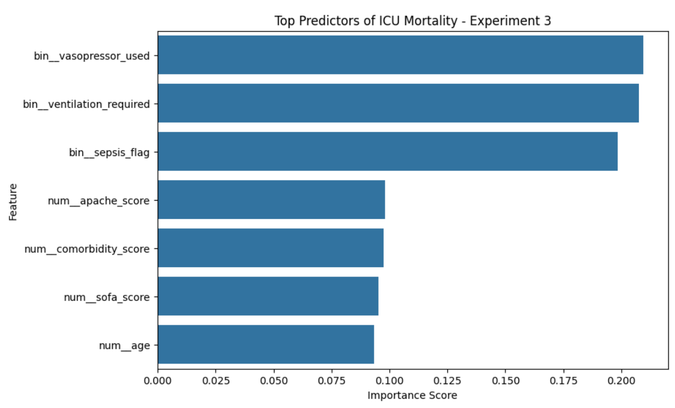
Key Visuals



This chart shows a clear class imbalance, with 11,582 survivors and 3,418 deaths. This imbalance can bias models toward predicting survival, making it important to apply techniques that improve detection of high-risk patients.

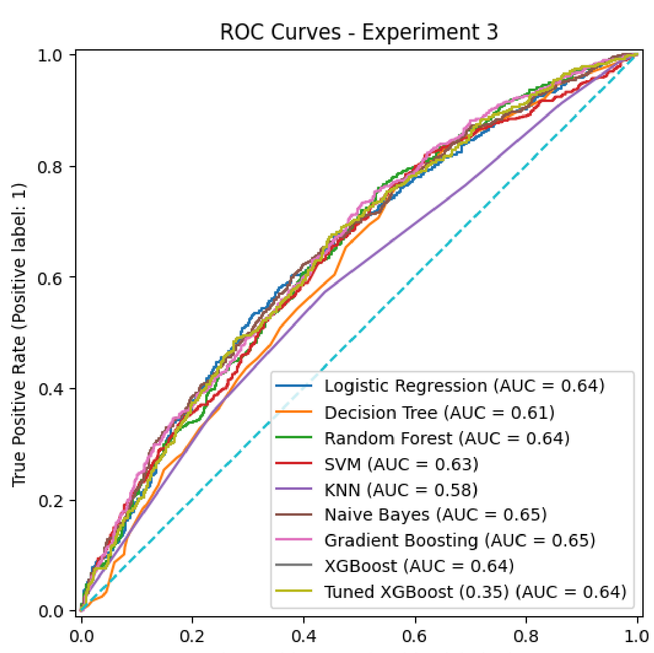


This matrix shows that the model correctly identifies many high-risk patients, achieving high recall, but also produces some false positives. This reflects the design choice to prioritize detecting critical cases over minimizing errors.



This graph highlights the most influential features in predicting mortality. Variables like vasopressor use, ventilation, and sepsis have the highest impact, followed by severity scores such as APACHE and SOFA, aligning with clinical expectations.

Key Visuals



This graph compares multiple models based on their ability to distinguish between survival and mortality. Most models have similar performance with AUC around 0.64–0.65, indicating moderate predictive ability. Tuned XGBoost performs competitively while supporting a recall-focused approach.

Results

The best-performing model was a tuned XGBoost classifier with a decision threshold of 0.35. The model achieved high recall, ensuring that most high-risk patients were correctly identified. Performance metrics included moderate accuracy and ROC-AUC, reflecting the complexity of the problem. The results demonstrate that prioritizing recall leads to better identification of critical cases, even if it introduces some false positives. This demonstrates that threshold tuning plays a key role in improving detection of high-risk ICU patients.

Final Model Summary

Final Model (Tuned XGBoost, threshold = 0.35)

- Accuracy: **58%**
- Precision: **55%**
- Recall: **88.45%**
- F1-score: **67.83%**
- ROC-AUC: **64%**

Analysis

The results show that ICU mortality prediction is strongly influenced by clinical indicators such as ventilation requirement, vasopressor use, and severity scores. The trade-off between recall and precision highlights the importance of aligning model performance with clinical priorities. Robustness testing revealed that noisy data increases false positives but can improve sensitivity, emphasizing the need to evaluate models under realistic conditions.

Conclusion

This project shows that machine learning can support early detection of high-risk ICU patients by prioritizing recall and handling class imbalance. The model provides a practical decision-support approach, though performance is moderate and sensitive to noisy data. Future work includes using real datasets and improving model robustness.

Limitation

- Moderate ROC-AUC performance
- Sensitive to noisy data
- Limited to one dataset

Key Sources & Acknowledgements

This work is based on publicly available ICU datasets and supported by prior research in healthcare machine learning and predictive modeling. We acknowledge the contributions of existing clinical scoring systems and machine learning frameworks that informed this study.